

Project Documentation

File: line simulation v3.project

Date: 04/07/2026

Profile: CODESYS V3.5 SP20 Patch 1

1 DUT: e_state

```
1  TYPE e_state :
2  (
3      OFF := 1 , // LINE IS OFF
4      HEATING := 2 , // OVEN IS HEATING, HEATING ELEMENT IS ON
5      WORKING := 3 , // THE OVEN IS IN WORKING TEMPTURE
6      UNLOADING := 4 , // THE LINE IS CLEARING PRODUCT
7      COOLING := 5 , // THE OVEN IS OVER HEATING HEATING ELEMENT IS OFF
8      CLEARING := 6 , //CLEARING PRODUCT FROM LINE
9      EMERGENCY := 7 // THE LINE IS STOPPED
10 ) ;
11 END_TYPE
12
```

2 Global Variable List: GVL

```
1  {attribute 'qualified_only'}
2  VAR_GLOBAL
3      iRawAnalogIn : INT ; // GLOBAL ANALOG INPUT OF RAW TEMPTURE
4      rTempCout : REAL ; // GLOBAL TEMPTURE SENSOR OUTPUT IN CELCIUS
5      bLocationsensor : BOOL ; // GLOBAL PRUDUCT LOCATION SENSOR
6  END_VAR
7
```

3 POU: fb_OvenState

```
1  FUNCTION_BLOCK fb_OvenState
2  VAR_INPUT
3      bSTART : BOOL ; // START LINE OPERATION
4      bSTOP : BOOL ; // STOP LINE OPERATION
5      bRESET : BOOL ; // RESET BUTTON TO CLEAR FAULTS AND EMERGENCY STATES
6      bProductLocation : BOOL ; // PRODUCT DETECTION SENSOR INSIDE THE OVEN
7      rTargetTemp : REAL := 200.0 ; // TARGET WORKING TEMPERATURE SETPOINT
8  END_VAR
9
10 VAR_IN_OUT
11     // FIX #5: CHANGED FROM VAR_INPUT TO VAR_IN_OUT TO ALLOW COMPLIANT
12     SIMULATION WRITING
13     rCorentTemp : REAL ; // CURRENT TEMPERATURE FEEDBACK FROM OVEN SENSOR
14     rRecordFbTemp : REAL ; // INPUT OF THE LAST RECORDED TEMP FROM THE
15     RECORD BLOCK
16 END_VAR
17
18 VAR_OUTPUT
19     eWorkstate : e_state ; // CURRENT MACHINE STATE BASED ON E_STATE ENUM
20     bHeatingElemnt : BOOL ; // HEATING ELEMENT CONTROL COMMAND (ON/OFF)
21     bFan : BOOL ; // OVEN COOLING FAN CONTROL COMMAND (ON/OFF)
22     bConveyorF : BOOL ; // CONVEYOR FORWARD MOVEMENT COMMAND
23     bConveyorB : BOOL ; // CONVEYOR BACKWARD MOVEMENT COMMAND
24     sMessage : STRING ; // SYSTEM STATUS MESSAGE FOR HMI DISPLAY
25 END_VAR
```

3 POU: fb_OvenState

```
24
25  VAR
26      // RISING EDGE TRIGGERS FOR DETECTING SINGLE BUTTON PRESS
27      rStop : R_TRIG ;
28      rReset : R_TRIG ;
29      rStart : R_TRIG ;
30      tHeatCheckTimer : TON ; // TIMER FOR CHECKING OVEN TEMPERATURE
31  STABILITY
32      tProsesTimer : TON ; // TIMER FOR PRODUCT PROCESSING DURATION
33      tOvenHeatingTimer : TON ; // HEATING TIMEOUT MONITORING TIMER
34      tTempCheckTimer : TON ; // TIMER FOR CHECKING IF COOLING IS WORKING
35
36      // INTERNAL CONTROL FLAGS TO PROPERLY TRIGGER AND RESET TIMERS OUTSIDE
37  THE CASE STRUCTURE
38      bTriggerHeatCheck : BOOL ;
39      bTriggerProses : BOOL ;
40      bTriggerOvenHeating : BOOL ;
41      bTriggerTempCheck : BOOL ; // TRIGGER FLAG FOR THE COOLING CHECK TIMER
42      bInit : BOOL := FALSE ; // FIRST SCAN INITIALIZATION FLAG
43      bSystemWasRunning : BOOL := FALSE ; // TRACKS WHETHER THE MACHINE WAS
44  ACTIVELY RUNNING
45
46      // INTERNAL CONTROL FLAGS FOR SAFETY AND PRODUCT SWAP MANAGEMENT
47      bSafetyOk : BOOL ; // SAFETY INTERLOCK FLAG (TRUE WHEN SYSTEM IS SAFE
48  TO RUN)
49      bProductUnloading : BOOL ; // HANDSHAKE FLAG TO TRACK COMPLETED PRODUCT
50  DISCHARGE
51
52      bStopLock : BOOL ; // LOCKS START ATTEMPTS AFTER STOP PRESS
53      bStartLatch : BOOL ; // START COMMAND LATCH FLAG TO PREVENT PULSE LOSS
54  IN SIMULATION
55      bCommandBlock : BOOL ; // GHOST EDGE LOCK FOR START FUNCTION
56  END_VAR
57
58  1 // FIRST SCAN INITIALIZATION
59  2 IF NOT bInit THEN
60  3     bSafetyOk := TRUE ;
61  4     bInit := TRUE ;
62  5 END_IF ;
63
64  6
65  7 // EVALUATE TRIGGER INPUTS NATURALLY (NO INTERFERENCE)
66  8 rStop ( CLK := bSTOP ) ;
67  9 rReset ( CLK := bRESET ) ;
68  10 rStart ( CLK := bSTART ) ;
69  11
70  12 // CAPTURE START COMMAND INTO INTERNAL LATCH
71  13 // CAPTURES THE EDGE PULSE AND HOLDS IT EVEN IF THE R_TRIG.Q DROPS ON THE
72  NEXT SCAN.
73  14 // IT WILL BE BLOCKED IF THE SYSTEM IS ACTIVELY LOCKED BY A STOP OR A
74  COMMAND BLOCK WINDOW.
75  15 IF rStart.Q AND bSafetyOk AND NOT bStopLock AND NOT bCommandBlock THEN
76  16     bStartLatch := TRUE ;
77  17 END_IF ;
78  18
79  19 // GLOBAL ROUTINE STOP INTERRUPT
```

3 POU: fb_OvenState

```
20      // STOP BUTTON IMMEDIATELY KILLS THE START LATCH AND LOCKS DOWN FURTHER
      COMMANDS.
21      IF rStop.Q AND eWorkstate <> e_state.OFF AND eWorkstate <> e_state.
      CLEARING AND eWorkstate <> e_state.EMERGENCY THEN
22          eWorkstate      := e_state.OFF ;
23          bStartLatch      := FALSE ; // FORCIBLY UNLATCH START REQUEST
24          bStopLock        := TRUE ; // ACTIVATE STATE LOCK
25          bCommandBlock    := TRUE ; // ACTIVATE COMMAND BLOCKING WINDOW
26
27          // FULL OPERATIONAL FLAGS RESET
28          bTriggerHeatCheck := FALSE ;
29          bTriggerOvenHeating := FALSE ;
30          bTriggerTempCheck := FALSE ;
31          bTriggerProses     := FALSE ;
32          bProductUnloading  := FALSE ;
33          bSystemWasRunning  := FALSE ;
34      END_IF
35
36      // GLOBAL RESET & UNLOCK HANDLER
37      // RESET BUTTON IS THE OFFICIAL WAY TO CLEAR THE LOCKS AND OPEN THE CLEAN
      COMMAND WINDOW.
38      IF rReset.Q AND eWorkstate = e_state.OFF THEN
39          bSafetyOk      := TRUE ;
40          bSystemWasRunning := FALSE ;
41          bStopLock      := FALSE ;
42          bCommandBlock  := FALSE ;
43          bStartLatch    := FALSE ; // ENSURE NO GHOST STARTUP HAPPENS UPON
      RESET
44      END_IF
45
46      // CRITICAL SAFETY TRIP INTERRUPT
47      // HARDWARE/PROCESS FAULTS PUSH THE SYSTEM TO EMERGENCY.
48      IF NOT bSafetyOk AND eWorkstate <> e_state.EMERGENCY THEN
49          bStartLatch := FALSE ; // INSTANTLY DROP START LATCH ON HARDWARE FAULT
50          eWorkstate  := e_state.EMERGENCY ;
51      END_IF
52
53      // MAIN STATE MACHINE LOGIC
54      CASE eWorkstate OF
55
56          e_state.OFF :
57              // HARD INTERLOCK - COMPONENTS MUST REMAIN INACTIVE IN OFF STATE
58              bHeatingElemnt := FALSE ;
59              bFan            := FALSE ;
60              bConveyorB     := FALSE ;
61              bConveyorF     := FALSE ;
62
63              // DYNAMIC MESSAGE CONTROL
64              IF bStopLock OR bCommandBlock THEN
65                  sMessage := 'SYSTEM LOCKED BY STOP - PRESS RESET TO UNLOCK
      COMMAND WINDOW' ;
66              ELSE
67                  sMessage := 'SYSTEM IS OFF - PRESS START TO ACTIVATE' ;
68              END_IF
69
70              // RESET TIMERS CONTROL FLAGS IN OFF
```

3 POU: fb_OvenState

```
71         bTriggerHeatCheck      := FALSE ;
72         bTriggerOvenHeating     := FALSE ;
73         bTriggerTempCheck       := FALSE ;
74         bTriggerProses           := FALSE ;
75         bProductUnloading       := FALSE ;
76
77         // HARD STATE CLEANUP FOR TON TIMERS
78         tHeatCheckTimer ( IN := FALSE ) ;
79         tOvenHeatingTimer ( IN := FALSE ) ;
80         tTempCheckTimer ( IN := FALSE ) ;
81         tProsesTimer ( IN := FALSE ) ;
82
83         // PRIORITY 1: PRODUCT DETECTED ON STARTUP -> GO TO CLEARING
84         IF bProductLocation = TRUE THEN
85             bStartLatch := FALSE ; // Abort start request if oven is
blocked
86             eWorkstate := e_state . CLEARING ;
87
88             // PRIORITY 2: AUTOMATIC ENVIRONMENTAL COOLING
89             ELIF bSystemWasRunning AND ( rCorentTemp > ( rTargetTemp + 15.0 )
) AND ( rCorentTemp > 50.0 ) THEN
90                 bStartLatch := FALSE ; // Abort start request if oven requires
cooling safety loop
91                 sMessage := 'OVEN TOO HOT! AUTOMATIC COOLING ACTIVATED' ;
92                 eWorkstate := e_state . COOLING ;
93
94                 // PRIORITY 3: ROUTINE START SEQUENCE (USING STABLE LATCH INTERFACE)
95                 // Evaluates bStartLatch instead of raw volatile rStart.Q pulse.
96                 ELIF bStartLatch AND bSafetyOk AND NOT bStopLock AND NOT
bCommandBlock AND NOT bSystemWasRunning THEN
97                     bSystemWasRunning := TRUE ;
98                     bStartLatch := FALSE ; // <<< CRITICAL HANDSHAKE: Reset
latch upon successful transition!
99                     eWorkstate := e_state . HEATING ;
100                 END_IF
101
102                 e_state . HEATING :
103                     bHeatingElemnt := TRUE ;
104                     bFan := FALSE ;
105                     bConveyorB := FALSE ;
106                     bConveyorF := FALSE ;
107                     sMessage := 'OVEN HEATING UP' ;
108
109                     bTriggerHeatCheck := TRUE ;
110                     bTriggerOvenHeating := TRUE ;
111
112                     // FAULT CHECK: TEMP NOT RISING
113                     IF tHeatCheckTimer . Q AND rCorentTemp > 50.0 THEN
114                         IF rCorentTemp <= rRecordFbTemp THEN
115                             bTriggerOvenHeating := FALSE ;
116                             bTriggerHeatCheck := FALSE ;
117                             bSafetyOk := FALSE ;
118                             sMessage := 'HEATING FAILURE! OVEN TEMPERATURE
IS NOT RISING' ;
119                             eWorkstate := e_state . EMERGENCY ;
120                         END_IF
```

```

121         END_IF
122
123         // FAULT CHECK: HEATING TIMEOUT
124         IF tOvenHeatingTimer.Q AND ( rCorentTemp < rTargetTemp ) THEN
125             bTriggerOvenHeating := FALSE ;
126             bTriggerHeatCheck   := FALSE ;
127             bSafetyOk           := FALSE ;
128             sMessage             := 'HEATING READ TIME OUT! EMERGENCY STATE'
129         ;
130         eWorkstate               := e_state . EMERGENCY ;
131         END_IF
132
133         // NORMAL TRANSITION
134         IF rCorentTemp >= rTargetTemp THEN
135             bTriggerOvenHeating := FALSE ;
136             bTriggerHeatCheck   := FALSE ;
137             eWorkstate          := e_state . WORKING ;
138         END_IF
139
140         e_state . WORKING :
141         // THERMAL RUNAWAY PROTECTION
142         IF rCorentTemp > ( rTargetTemp + 15.0 ) THEN
143             bTriggerProses := FALSE ;
144             eWorkstate     := e_state . COOLING ;
145         END_IF
146
147         // CRASH DETECTION
148         IF rCorentTemp < 40.0 THEN
149             bTriggerProses := FALSE ;
150             bSafetyOk      := FALSE ;
151             sMessage       := 'CRITICAL SENSOR OR ELEMENT FAILURE
152 DETECTED!' ;
153             eWorkstate     := e_state . EMERGENCY ;
154         END_IF
155
156         // PROCESS STEP LOGIC
157         IF eWorkstate = e_state . WORKING THEN
158             IF bProductLocation = TRUE THEN
159                 bConveyorF      := FALSE ;
160                 bTriggerProses   := TRUE ;
161                 sMessage         := 'PRODUCT IN PROCESS' ;
162                 bHeatingElemnt   := ( rCorentTemp < rTargetTemp ) ;
163
164                 IF tProsesTimer.Q THEN
165                     bTriggerProses := FALSE ;
166                     IF ( rCorentTemp >= ( rTargetTemp - 15.0 ) ) AND (
167 rCorentTemp <= ( rTargetTemp + 15.0 ) ) THEN
168                         sMessage := 'PROCESS SUCCESSFUL - ROUTING
169 TO UNLOAD' ;
170                         bProductUnloading := TRUE ;
171                         eWorkstate := e_state . UNLOADING ;
172                     ELSE
173                         sMessage := 'PROCESS TEMPERATURE FAULT! REJECTING
174 ITEM' ;
175                         eWorkstate := e_state . CLEARING ;
176                     END_IF

```

3 POU: fb_OvenState

```
172         END_IF
173     ELSE
174         IF NOT bTriggerProses THEN
175             bConveyorF      := TRUE ;
176             bHeatingElemnt  := ( rCorentTemp < rTargetTemp ) ;
177             sMessage        := 'LOADING NEW PRODUCT' ;
178         END_IF
179     END_IF
180 END_IF
181
182 e_state . UNLOADING :
183     bConveyorF      := TRUE ;
184     bHeatingElemnt  := ( rCorentTemp < rTargetTemp ) ;
185     bTriggerProses  := FALSE ;
186     sMessage        := 'PRODUCT SWAP IN PROGRESS - WAITING FOR ENTRY' ;
187
188     IF NOT bProductLocation THEN
189         bProductUnloading := FALSE ;
190         sMessage          := 'PREVIOUS PRODUCT CLEARED - WAITING FOR
191 NEXT' ;
192     END_IF
193
194     IF NOT bProductUnloading AND bProductLocation = TRUE THEN
195         sMessage      := 'NEW PRODUCT DETECTED - RESTARTING PROCESS' ;
196         eWorkstate    := e_state . WORKING ;
197     END_IF
198
199     IF rCorentTemp > ( rTargetTemp + 15.0 ) THEN
200         eWorkstate := e_state . COOLING ;
201     END_IF
202
203 e_state . COOLING :
204     bHeatingElemnt  := FALSE ;
205     bFan            := TRUE ;
206     bConveyorF      := FALSE ;
207     bTriggerTempCheck := TRUE ;
208     sMessage        := 'COOLING IN PROGRESS, REJECTING PRODUCT' ;
209
210     IF bProductLocation = TRUE THEN
211         bConveyorB := TRUE ;
212     ELSE
213         bConveyorB := FALSE ;
214     END_IF
215
216     // TREND PROTECTION FOR COOLING
217     IF tTempCheckTimer . Q AND rCorentTemp > 50.0 THEN
218         IF rCorentTemp >= rRecordFbTemp THEN
219             bTriggerTempCheck := FALSE ;
220             bSafetyOk         := FALSE ;
221             sMessage          := 'COOLING FAILURE DETECTED! OVEN
222 TEMPERATURE IS NOT DROPPING' ;
223             eWorkstate        := e_state . EMERGENCY ;
224         END_IF
225     END_IF
226
227     // COOLING EXIT CONDITIONS
```

```

226         IF eWorkstate = e_state.COOLING THEN
227             IF ( rCorentTemp > ( rTargetTemp + 30.0 ) ) THEN
228                 bTriggerTempCheck := FALSE ;
229                 bSafetyOk          := FALSE ;
230                 sMessage           := 'CRITICAL THERMAL ESCALATION! HARD
LOCKED' ;
231                 eWorkstate         := e_state.EMERGENCY ;
232
233                 ELSIF rCorentTemp > ( rTargetTemp + 15.0 ) THEN
234                     sMessage := 'CRITICAL COOLING ACTIVE' ;
235                 ELSEIF rCorentTemp <= rTargetTemp THEN
236                     bTriggerTempCheck := FALSE ;
237                     bSystemWasRunning := FALSE ;
238                     sMessage := 'COOLING COMPLETE - SYSTEM SAFE' ;
239                     eWorkstate := e_state.OFF ;
240                 END_IF
241             END_IF
242             e_state.CLEARING :
243             bHeatingElemnt := FALSE ;
244             bFan := FALSE ;
245             bConveyorF := FALSE ;
246             sMessage := 'CLEARING TRAPPED PRODUCT - CONVEYOR REVERSING' ;
247             IF bProductLocation = TRUE THEN
248                 bConveyorB := TRUE ;
249             ELSE
250                 bConveyorB := FALSE ;
251                 eWorkstate := e_state.OFF ;
252             END_IF
253             e_state.EMERGENCY :
254             bHeatingElemnt := FALSE ;
255             bFan := FALSE ;
256             bConveyorB := FALSE ;
257             bConveyorF := FALSE ;
258             bTriggerTempCheck := FALSE ;
259             bTriggerHeatCheck := FALSE ;
260             sMessage := 'CRITICAL FAULT! PRESS RESET TO CLEAR TRAPPED ERROR' ;
261             IF rReset.Q THEN
262                 bStopLock := FALSE ;
263                 bCommandBlock := FALSE ;
264                 IF rCorentTemp > rTargetTemp THEN
265                     eWorkstate := e_state.COOLING ;
266                 ELSE
267                     bSafetyOk := TRUE ;
268                     bSystemWasRunning := FALSE ;
269                     eWorkstate := e_state.OFF ;
270                 END_IF
271             END_IF
272         END_CASE
273
274
275         // EXECUTE ALL INTERNAL TIMERS OUTSIDE THE CASE STRUCTURE ---
276         tHeatCheckTimer ( IN := bTriggerHeatCheck , PT := T#5S ) ;
277         tOvenHeatingTimer ( IN := bTriggerOvenHeating , PT := T#15S ) ;
278         tTempCheckTimer ( IN := bTriggerTempCheck , PT := T#20S ) ;
279         tProsesTimer ( IN := bTriggerProses , PT := T#10S ) ;
280

```

```

281
282 // INTERNAL OVEN SIMULATION LOGIC ---
283 // Simulates live physical feedback based on active outputs
284 IF bHeatingElemnt THEN
285     rCorentTemp := rCorentTemp + 0.15 ;
286 ELSEIF bFan THEN
287     rCorentTemp := rCorentTemp - 0.25 ;
288 ELSE
289     IF rCorentTemp > 25.0 THEN
290         rCorentTemp := rCorentTemp - 0.01 ; // Slow environmental heat
loss
291     END_IF
292 END_IF
293
294
295 // STANDARD COMPLIANT TREND TRACKING FOR SIMULATION ---
296 // Captures a snapshot of the temperature when the check begins, allowing
the trend logic
297 // in HEATING and COOLING states to evaluate a real mathematical slope.
298 IF eWorkstate = e_state.HEATING AND bTriggerHeatCheck AND NOT
tHeatCheckTimer.Q THEN
299     // If the timer just started, sample the baseline temperature once
300     IF tHeatCheckTimer.ET = T#0S THEN
301         rRecordFbTemp := rCorentTemp ;
302     END_IF
303 ELSEIF eWorkstate = e_state.COOLING AND bTriggerTempCheck AND NOT
tTempCheckTimer.Q THEN
304     // If the cooling timer just started, sample the baseline temperature
once
305     IF tTempCheckTimer.ET = T#0S THEN
306         rRecordFbTemp := rCorentTemp ;
307     END_IF
308 END_IF
309

```

4 POU: fb_sensorRcored

```

1 FUNCTION_BLOCK fb_sensorRcored
2 VAR_INPUT
3     rCorentLogTemp : REAL ; // CURRENT TEMPERATURE FEEDBACK TO BE LOGGED
4     tLogInterval : TIME := T#20S ; // TIME CYCLE FOR EACH LOG ENTRY
    (DEFAULT 20 SECONDS)
5 END_VAR
6 VAR_OUTPUT
7     // AN ARRAY STORES THE LAST 10 LOGGED TEMPERATURE VALUES FOR SIMULATION
TRACKING
8     aTempHistory : ARRAY [ 1 .. 10 ] OF REAL ;
9 END_VAR
10 VAR
11     tLogTimer : TON ; // INTERNAL TIMER TO TRIGGER THE LOG EVERY 20 SECONDS
12     i : INT ; // LOOP INDEX FOR MOVING ARRAY DATA
13     rLastStoredTemp : REAL ;
14 END_VAR
15

```

4 POU: fb_sensorRcored

```
1 // EXECUTE THE REPEATING LOG TIMER WITH AUTOMATIC RESTART TRIGGER
2 tLogTimer ( IN := NOT tLogTimer.Q, PT := tLogInterval );
3
4 // EVERY TIME THE TIMER REACHES 20 SECONDS, TRIGGER THE SHIFT AND RECORD
  SEQUENCE
5 IF tLogTimer.Q THEN
6
7     // SHIFT ALL EXISTING LOGGED VALUES ONE SLOT DOWN IN THE ARRAY FROM
  POSITION 1 UP TO 9
8     FOR i := 1 TO 9 BY 1 DO
9         aTempHistory [ i ] := aTempHistory [ i + 1 ];
10    END_FOR ;
11
12    // INSERT THE LATEST CURRENT TEMPERATURE VALUE INTO THE NEWEST POSITION
  SLOT 10
13    aTempHistory [ 10 ] := rCorentLogTemp ;
14
15    // UPDATE THE OUTPUT WITH THE RECENTLY RECORDED VALUE TO SEND IT TO THE
  OVEN STATE MACHINE
16    rLastStoredTemp := rCorentLogTemp ;
17
18 END_IF
19
20 // INITIALIZATION PROTECTION: IF THE LOGGED TEMP IS STILL ZERO DURING THE
  FIRST 20 SECONDS, CAPTURE CURRENT TEMP AS BASELINE
21 IF rLastStoredTemp = 0.0 THEN
22     rLastStoredTemp := rCorentLogTemp ;
23 END_IF
24
```

5 POU: fb_SensorScaling

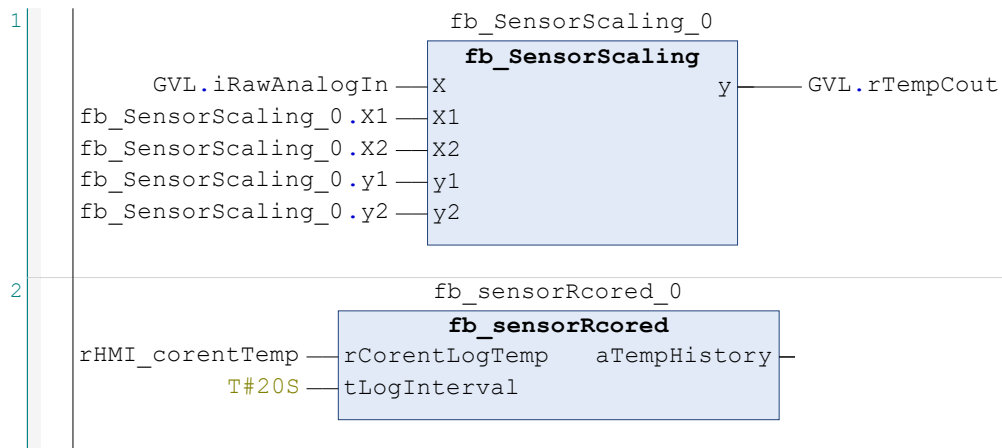
```
1 FUNCTION_BLOCK fb_SensorScaling
2 VAR_INPUT
3     X : INT ; // INPUT OF iRawAnalogIn SENSOR TO SCALING FUNCTION
4     X1 : REAL := 0.0 ; // LOWER LIMIT OF THE SENSOR
5     X2 : REAL := 27648.0 ; // UPER LIMIT OF THE SENSOR BASED ON SIEMNES PLC
6     y1 : REAL := 23.0 ; // LOWER OUTPUT RANGE FOR SENSOR TO SIMOLATE ROOM
  TEMPTURE
7     y2 : REAL := 400.0 ; // UPER OUTPOT RANGE FOR SENSOR IN CELCIUS
8 END_VAR
9 VAR_OUTPUT
10    y : REAL ; // OUTPUT OF rTempCout SENSOR FROM SCALING FUNCTION
11 END_VAR
12 VAR
13 END_VAR
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
105
106
107
108
109
110
111
112
113
114
115
116
117
118
119
120
121
122
123
124
125
126
127
128
129
130
131
132
133
134
135
136
137
138
139
140
141
142
143
144
145
146
147
148
149
150
151
152
153
154
155
156
157
158
159
160
161
162
163
164
165
166
167
168
169
170
171
172
173
174
175
176
177
178
179
180
181
182
183
184
185
186
187
188
189
190
191
192
193
194
195
196
197
198
199
200
201
202
203
204
205
206
207
208
209
210
211
212
213
214
215
216
217
218
219
220
221
222
223
224
225
226
227
228
229
230
231
232
233
234
235
236
237
238
239
240
241
242
243
244
245
246
247
248
249
250
251
252
253
254
255
256
257
258
259
260
261
262
263
264
265
266
267
268
269
270
271
272
273
274
275
276
277
278
279
280
281
282
283
284
285
286
287
288
289
290
291
292
293
294
295
296
297
298
299
300
301
302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
327
328
329
330
331
332
333
334
335
336
337
338
339
340
341
342
343
344
345
346
347
348
349
350
351
352
353
354
355
356
357
358
359
360
361
362
363
364
365
366
367
368
369
370
371
372
373
374
375
376
377
378
379
380
381
382
383
384
385
386
387
388
389
390
391
392
393
394
395
396
397
398
399
400
401
402
403
404
405
406
407
408
409
410
411
412
413
414
415
416
417
418
419
420
421
422
423
424
425
426
427
428
429
430
431
432
433
434
435
436
437
438
439
440
441
442
443
444
445
446
447
448
449
450
451
452
453
454
455
456
457
458
459
460
461
462
463
464
465
466
467
468
469
470
471
472
473
474
475
476
477
478
479
480
481
482
483
484
485
486
487
488
489
490
491
492
493
494
495
496
497
498
499
500
501
502
503
504
505
506
507
508
509
510
511
512
513
514
515
516
517
518
519
520
521
522
523
524
525
526
527
528
529
530
531
532
533
534
535
536
537
538
539
540
541
542
543
544
545
546
547
548
549
550
551
552
553
554
555
556
557
558
559
560
561
562
563
564
565
566
567
568
569
570
571
572
573
574
575
576
577
578
579
580
581
582
583
584
585
586
587
588
589
590
591
592
593
594
595
596
597
598
599
600
601
602
603
604
605
606
607
608
609
610
611
612
613
614
615
616
617
618
619
620
621
622
623
624
625
626
627
628
629
630
631
632
633
634
635
636
637
638
639
640
641
642
643
644
645
646
647
648
649
650
651
652
653
654
655
656
657
658
659
660
661
662
663
664
665
666
667
668
669
670
671
672
673
674
675
676
677
678
679
680
681
682
683
684
685
686
687
688
689
690
691
692
693
694
695
696
697
698
699
700
701
702
703
704
705
706
707
708
709
710
711
712
713
714
715
716
717
718
719
720
721
722
723
724
725
726
727
728
729
730
731
732
733
734
735
736
737
738
739
740
741
742
743
744
745
746
747
748
749
750
751
752
753
754
755
756
757
758
759
760
761
762
763
764
765
766
767
768
769
770
771
772
773
774
775
776
777
778
779
780
781
782
783
784
785
786
787
788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809
810
811
812
813
814
815
816
817
818
819
820
821
822
823
824
825
826
827
828
829
830
831
832
833
834
835
836
837
838
839
840
841
842
843
844
845
846
847
848
849
850
851
852
853
854
855
856
857
858
859
860
861
862
863
864
865
866
867
868
869
870
871
872
873
874
875
876
877
878
879
880
881
882
883
884
885
886
887
888
889
890
891
892
893
894
895
896
897
898
899
900
901
902
903
904
905
906
907
908
909
910
911
912
913
914
915
916
917
918
919
920
921
922
923
924
925
926
927
928
929
930
931
932
933
934
935
936
937
938
939
940
941
942
943
944
945
946
947
948
949
950
951
952
953
954
955
956
957
958
959
960
961
962
963
964
965
966
967
968
969
970
971
972
973
974
975
976
977
978
979
980
981
982
983
984
985
986
987
988
989
990
991
992
993
994
995
996
997
998
999
1000
```

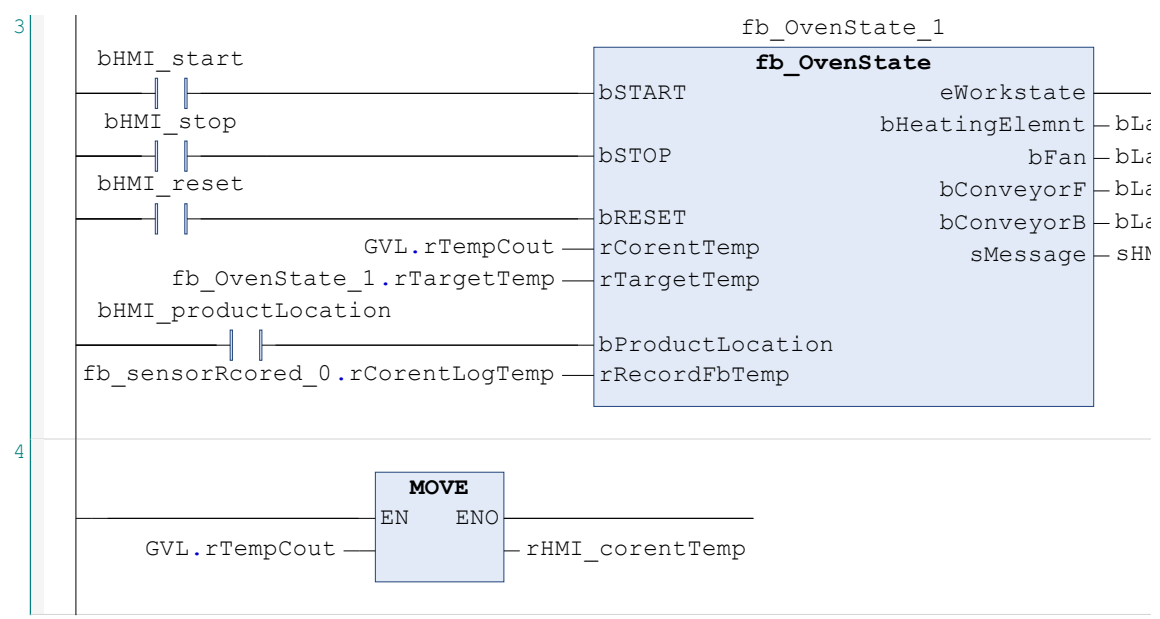
6 POU: PLC_PRG

```

1  PROGRAM PLC_PRG
2  VAR
3      fb_SensorScaling_0 : fb_SensorScaling ; // DEFNATION OF THE TEMTURE
        SCALING FUNCTIONAL BLOCK
4      fb_OvenState_1 : fb_OvenState ; //DEFNATION OF THE OVEN STATE FUNCTIONAL
        BLOCK
5      fb_sensorRcored_0 : fb_sensorRcored ; // DEFNATION OF THE RECOREDING
        SENSOR BLOCK
6      // LINKING INPUTS
7      bHMI_start : BOOL ;
8      bHMI_stop : BOOL ;
9      bHMI_reset : BOOL ; // NEW: HMI RESET BUTTON TO CLEAR EMERGENCIES AND
        SAFETY FAULTS
10     rHMI_corentTemp : REAL ;
11     rHMI_targetTemp : REAL := 200.0 ;
12     bHMI_productLocation : BOOL ;
13
14     // LINKING OUTPUTS
15     bLamp_heating : BOOL ;
16     bLamp_Fan : BOOL ;
17     bLamp_ConveyorF : BOOL ;
18     bLamp_Conveyorb : BOOL ;
19     bLamp_Reset : BOOL ; // NEW: RESET PROMPT LAMP (FLASHES WHEN RESET IS
        REQUIRED)
20     sHMI_Message : STRING ;
21
22 END_VAR
23

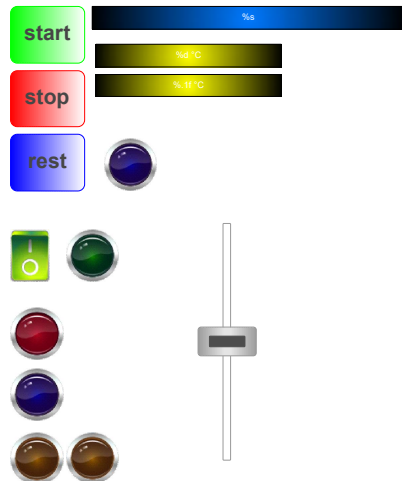
```





amp_heating
amp_Fan
amp_ConveyorF
amp_Conveyorb
MI_Message

7 Visualization: Visualization



General

Visualization size algorithm version: Respecting scrollbar location

Background

Use background color: False

Background color: 16777215

Interface

VAR_IN_OUT

END_VAR

Visualization Element List

Button Id: 0

Element name: GenElemInst_2

Type of element: Button

Tab Order: default

Static optimized: False

Text ID: 467

Position

X: 28

Y: 23

Width: 98

Height: 75

Center

X: 77

Y: 60

Colors

Color

Transparency: 255

Alarm color

Transparency: 255

Use gradient color: True

Gradient setting

Gradient

Color1: Green
Color2: Color [A=255, R=255, G=255, B=255]
Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Button height: 0
Bitmap info
Scaling type: Isotropic
Transparent: False
Transparent color: Black
Horizontal alignment: Left
Vertical alignment: Top
Texts
Text: start
Text properties
Horizontal alignment: Centered
Vertical alignment: Centered
Text format: Default
Font: Arial;28;Bold;Element-Button-FontColor, Color [A=255, R=0, G=0, B=0]
Font color
Transparency: 255
Font variables
Size: , <pt>
Button state variable
Digital variable: <NOT toggle/tap variable>
Animation duration: 0
Input configuration
OnDialogClosed: Configure...
OnMouseClicked: Configure...
OnMouseDown: Configure...
OnMouseEnter: Configure...
OnMouseLeave: Configure...
OnMouseMove: Configure...
OnMouseUp: Configure...
OnValueChanged: Configure...
Toggle
Toggle on up if captured: False
Tap
Variable: PLC_PRG.bHMI_start
Tap FALSE: True
Tap on enter if captured: False
■Button Id: 3
Element name: GenElemInst_3
Type of element: Button
Tab Order: default
Static optimized: False
Text ID: 514
Position
X: 28
Y: 108
Width: 98
Height: 75
Center
X: 77
Y: 145
Colors

Color
 Transparency: 255
Alarm color
 Transparency: 255
Use gradient color: True
Gradient setting
 Gradient
 Color1: Red
 Color2: Color [A=255, R=255, G=255, B=255]
 Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Button height: 0
Bitmap info
 Scaling type: Isotropic
 Transparent: False
 Transparent color: Black
 Horizontal alignment: Left
 Vertical alignment: Top
Texts
 Text: stop
Text properties
 Horizontal alignment: Centered
 Vertical alignment: Centered
 Text format: Default
 Font: Arial;28;Bold;Element-Button-FontColor, Color [A=255, R=0, G=0, B=0]
 Font color
 Transparency: 255
Font variables
 Size: , <pt>
Button state variable
 Digital variable: <NOT toggle/tap variable>
Animation duration: 0
Input configuration
 OnDialogClosed: Configure...
 OnMouseClicked: Configure...
 OnMouseDown: Configure...
 OnMouseEnter: Configure...
 OnMouseLeave: Configure...
 OnMouseMove: Configure...
 OnMouseUp: Configure...
 OnValueChanged: Configure...
Toggle
 Toggle on up if captured: False
Tap
 Variable: PLC_PRG.bHMI_stop
 Tap FALSE: True
 Tap on enter if captured: False
■ RockerSwitch Id: 5
 Element name: GenElemInst_5
 Type of element: RockerSwitch
 Tab Order: default
 Static optimized: False
Position
 X: 28
 Y: 319

Width: 70
Height: 70
Variable: PLC_PRG.bHMI_productLocation
Image settings
Transparent: True
Transparent color: BasicElement-Fill-Color
Isotropic type: Isotropic
Horizontal alignment: Left
Vertical alignment: Top
Element behavior: Image toggler
Tap FALSE: False
Center
X: 63
Y: 354

Animation duration: 0

Input configuration

OnValueChanged: Configure...

Background

Image: Green

● Lamp1 Id: 7

Element name: GenElemInst_7

Type of element: Lamp1

Tab Order: default

Static optimized: False

Position

X: 28

Y: 423

Width: 70

Height: 70

Variable: PLC_PRG.bLamp_heating

Image settings

Transparent: False

Transparent color: Red

Isotropic type: Isotropic

Horizontal alignment: Left

Vertical alignment: Top

Center

X: 63

Y: 458

Animation duration: 0

Background

Image: Red

● Lamp1 Id: 11

Element name: GenElemInst_9

Type of element: Lamp1

Tab Order: default

Static optimized: False

Position

X: 28

Y: 503

Width: 70

Height: 70

Variable: PLC_PRG.bLamp_Fan

Image settings

Transparent: False
Transparent color: Blue
Isotropic type: Isotropic
Horizontal alignment: Left
Vertical alignment: Top
Center
X: 63
Y: 538
Animation duration: 0
Background
Image: Blue
● Lamp1 Id: 13
Element name: GenElemInst_10
Type of element: Lamp1
Tab Order: default
Static optimized: False
Position
X: 28
Y: 588
Width: 70
Height: 70
Variable: PLC_PRG.bLamp_Conveyorb
Image settings
Transparent: False
Transparent color: Yellow
Isotropic type: Isotropic
Horizontal alignment: Left
Vertical alignment: Top
Center
X: 63
Y: 623
Animation duration: 0
Background
Image: Yellow
● Lamp1 Id: 15
Element name: GenElemInst_11
Type of element: Lamp1
Tab Order: default
Static optimized: False
Position
X: 102
Y: 588
Width: 70
Height: 70
Variable: PLC_PRG.bLamp_ConveyorF
Image settings
Transparent: False
Transparent color: Yellow
Isotropic type: Isotropic
Horizontal alignment: Left
Vertical alignment: Top
Center
X: 137
Y: 623

Animation duration: 0
Background
Image: Yellow
● Lamp1 Id: 17
Element name: GenElemInst_12
Type of element: Lamp1
Tab Order: default
Static optimized: False
Position
X: 102
Y: 319
Width: 70
Height: 70
Variable: PLC_PRG.bHMI_productLocation
Image settings
Transparent: False
Transparent color: Green
Isotropic type: Isotropic
Horizontal alignment: Left
Vertical alignment: Top
Center
X: 137
Y: 354
Animation duration: 0
Background
Image: Green
■ Rectangle Id: 19
Element name: GenElemInst_14
Tab Order: default
Static optimized: False
Text ID: 887
Type of element: Rectangle
Position
X: 136
Y: 23
Width: 415
Height: 30
Angle: 0
Radius setting
Radius: From style
Value: 5
Center
X: 343
Y: 38
Colors
Normal state
Frame color
Transparency: 255
Fill color
Transparency: 255
Alarm state
Frame color
Transparency: 255
Fill color

Transparency: 255
Use gradient color: True
Gradient setting
Gradient
Color1: Element-XYChart-Secondary-CursorColor
Color2: Color [A=255, R=255, G=255, B=255]
Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Appearance
Line width: 1
Fill attributes: Filled
Line style: Solid
Texts
Text: %s
Text properties
Horizontal alignment: Centered
Vertical alignment: Centered
Text format: Default
Font: Arial;12;Regular;Element-Background-Color, Color [A=255, R=255, G=255, B=255]
Font color
Transparency: 255
Absolute movement
Use REAL values: False
Text variables
Text variable: PLC_PRG.sHMI_Message
Font variables
Size: , <pt>
Color variables
Toggle color: <toggle/tap variable>
Animation duration: 0
Input configuration
OnDialogClosed: Configure...
OnMouseClicked: Configure...
OnMouseDown: Configure...
OnMouseEnter: Configure...
OnMouseLeave: Configure...
OnMouseMove: Configure...
OnMouseUp: Configure...
OnValueChanged: Configure...
Toggle
Toggle on up if captured: False
Tap
Tap FALSE: False
Tap on enter if captured: False
■Rectangle Id: 21
Element name: GenElemInst_16
Tab Order: default
Static optimized: False
Text ID: 960
Type of element: Rectangle
Position
X: 142
Y: 73
Width: 245
Height: 30

Angle: 0
Radius setting
 Radius: From style
 Value: 5
Center
 X: 264
 Y: 88
Colors
 Normal state
 Frame color
 Transparency: 255
 Fill color
 Transparency: 255
 Alarm state
 Frame color
 Transparency: 255
 Fill color
 Transparency: 255
Use gradient color: True
Gradient setting
 Gradient
 Color1: Yellow
 Color2: Color [A=255, R=255, G=255, B=255]
 Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Appearance
 Line width: 1
 Fill attributes: Filled
 Line style: Solid
Texts
 Text: %d °C
Text properties
 Horizontal alignment: Centered
 Vertical alignment: Centered
 Text format: Default
 Font: Arial;12;Regular;Element-Background-Color, Color [A=255, R=255, G=255, B=255]
 Font color
 Transparency: 255
Absolute movement
 Use REAL values: False
Text variables
 Text variable: PLC_PRG.rHMI_targetTemp
Font variables
 Size: , <pt>
Color variables
 Toggle color: <toggle/tap variable>
Animation duration: 0
Input configuration
 OnDialogClosed: Configure...
 OnMouseClicked
 Input Action
 ✱Write Variable
 Type: VisuDialogs.Numpad
 Variable: <Text output variable>
 Password: False

Dialog position type: UseGlobalSetting
OnMouseDown: Configure...
OnMouseEnter: Configure...
OnMouseLeave: Configure...
OnMouseMove: Configure...
OnMouseUp: Configure...
OnValueChanged: Configure...
Toggle
Toggle on up if captured: False
Tap
Tap FALSE: False
Tap on enter if captured: False
■Rectangle Id: 25
Element name: GenElemInst_19
Tab Order: default
Static optimized: False
Text ID: 263
Type of element: Rectangle
Position
X: 142
Y: 113
Width: 245
Height: 30
Angle: 0
Radius setting
Radius: From style
Value: 5
Center
X: 264
Y: 128
Colors
Normal state
Frame color
Transparency: 255
Fill color
Transparency: 255
Alarm state
Frame color
Transparency: 255
Fill color
Transparency: 255
Use gradient color: True
Gradient setting
Gradient
Color1: Yellow
Color2: Color [A=255, R=255, G=255, B=255]
Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Appearance
Line width: 1
Fill attributes: Filled
Line style: Solid
Texts
Text: %.1f °C
Text properties

Horizontal alignment: Centered
Vertical alignment: Centered
Text format: Default
Font: Arial;12;Regular;Element-Background-Color, Color [A=255, R=255, G=255, B=255]
Font color
Transparency: 255
Absolute movement
Use REAL values: False
Text variables
Text variable: PLC_PRG.rHMI_corentTemp
Font variables
Size: , <pt>
Color variables
Toggle color: <toggle/tap variable>
Animation duration: 0
Input configuration
OnDialogClosed: Configure...
OnMouseClicked: Configure...
OnMouseDown: Configure...
OnMouseEnter: Configure...
OnMouseLeave: Configure...
OnMouseMove: Configure...
OnMouseUp: Configure...
OnValueChanged: Configure...
Toggle
Toggle on up if captured: False
Tap
Tap FALSE: False
Tap on enter if captured: False
Slider Id: 27
Element name: GenElemInst_21
Type of element: Slider
Tab Order: default
Static optimized: False
Position
X: 277
Y: 292
Width: 77
Height: 352
Variable: GVL.iRawAnalogIn
Move to click: False
Scale
Show scale: False
Scale start: 0
Scale end: 27648
Main scale: 10
Subscale: 5
Scale format (C syntax): %i
Scale proportion: 0
Bar
Diagram type: Bottom
Diagram type: Right
Orientation: Vertical
Running direction: Left to right

Running direction: Top to bottom
Center
X: 315
Y: 468
Animation duration: 0
Input configuration
OnValueChanged: Configure...
■ Button Id: 30
Element name: GenElemInst_22
Type of element: Button
Tab Order: default
Static optimized: False
Text ID: 749
Position
X: 28
Y: 193
Width: 98
Height: 75
Center
X: 77
Y: 230
Colors
Color
Transparency: 255
Alarm color
Transparency: 255
Use gradient color: True
Gradient setting
Gradient
Color1: Blue
Color2: Color [A=255, R=255, G=255, B=255]
Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Button height: 0
Bitmap info
Scaling type: Isotropic
Transparent: False
Transparent color: Black
Horizontal alignment: Left
Vertical alignment: Top
Texts
Text: rest
Text properties
Horizontal alignment: Centered
Vertical alignment: Centered
Text format: Default
Font: Arial;28;Bold;Element-Button-FontColor, Color [A=255, R=0, G=0, B=0]
Font color
Transparency: 255
Font variables
Size: , <pt>
Button state variable
Digital variable: <NOT toggle/tap variable>
Animation duration: 0
Input configuration

OnDialogClosed: Configure...
OnMouseClicked: Configure...
OnMouseDown: Configure...
OnMouseEnter: Configure...
OnMouseLeave: Configure...
OnMouseMove: Configure...
OnMouseUp: Configure...
OnValueChanged: Configure...
Toggle
 Toggle on up if captured: False
Tap
 Variable: PLC_PRG.bHMI_reset
 Tap FALSE: True
 Tap on enter if captured: False
● Lamp1 Id: 32
 Element name: GenElemInst_23
 Type of element: Lamp1
 Tab Order: default
 Static optimized: False
 Position
 X: 153
 Y: 198
 Width: 70
 Height: 70
 Variable: PLC_PRG.bLamp_Reset
 Image settings
 Transparent: False
 Transparent color: Blue
 Isotropic type: Isotropic
 Horizontal alignment: Left
 Vertical alignment: Top
 Center
 X: 188
 Y: 233
 Animation duration: 0
 Background
 Image: Blue